

Stony Brook University
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MIDTERM I PRACTICE PROBLEMS

Due: Optional, but you should use it to prepare for the midterm.

Instructions: This is an optional assignment whose purpose is to prepare you for the midterm. I strongly recommend that you try to solve or at least understand all the questions.

1. Michele, who has a relatively high income I , has altruistic feelings toward Sofia, who lives in such poverty that she essentially has no income. Suppose Michele's preferences are represented by the utility function

$$U_M(c_M, c_S) = c_M^{1-\alpha} c_S^\alpha,$$

where c_M and c_S are Michele and Sofia's consumption levels, appearing as goods in a standard Cobb-Douglas utility function. Assume that Michele can spend her income either on her own or Sofia's consumption (through charitable donations) and that \$1 buys a unit of consumption for either (thus, the "prices" of consumption are $p_M = p_S = 1$).

- (a) Argue that the exponent α can be taken as a measure of the degree of Michele's altruism by providing an interpretation of extreme values of $\alpha = 0$ and $\alpha = 1$. What value would make her a perfect altruist (regarding others the same as oneself)?
 - (b) Solve for Michele's optimal choices and demonstrate how they change with α .
 - (c) Suppose that there is an income tax at rate τ , i.e. net income now is just $(1 - \tau)I$. Solve for Michele's optimal choices under the income tax rate.
 - (d) Now suppose that besides the income tax rate τ , there are charitable deductions, so that income spent on charitable deductions is not taxed. Argue that this amounts to changing the price p_S from \$1 to $\$(1 - \tau)$. Solve for the optimal choices under both the income tax rate and charitable deductions. Does the charitable deduction have a bigger incentive effect on more or less altruistic people?
2. Suppose that a fast-food junkie derives utility from three goods—soft drinks (x), hamburgers (y), and ice cream sundaes (z)—according to the utility function

$$U(x, y, z) = x^{0.5} y^{0.5} (1 + z)^{0.5}.$$

Suppose also that the prices for these goods are given by $p_x = 1$, $p_y = 4$, and $p_z = 8$ and that this consumer's income is given by $I = 8$.

- (a) Show that, for $z = 0$, maximization of utility results in the same optimal choices as in the case of a Cobb-Douglas utility function $U(x, y) = x^{0.5} y^{0.5}$. Show also that any choice that results in $z > 0$ (even for a fractional z) reduces utility from this optimum.
- (b) How do you explain the fact that $z = 0$ is optimal here?
- (c) How high would this individual's income have to be for any z to be purchased?

3. Consider the utility function

$$u(x, y) = (x + 2)(y + 3), \quad x \geq 0, y \geq 0$$

with the accompanying budget constraint:

$$p_x x + p_y y \leq I, \quad p_x, p_y, I > 0.$$

- Fix a given utility level $U > 0$ and find an explicit expression for the indifference curve defined by the utility level $U > 0$. Then, derive an explicit expression for the marginal rate of substitution between good x and good y .
- Draw the indifference curve (for this associated level of utility U) and carefully label the graph and its elements.
- Show that the utility function is strictly increasing (and hence monotone) in x and y .
- Now formally state the utility maximization problem and briefly describe its content, in particular what constitutes the choice variables, and what constitutes parameters.
- Provide an argument why in the present utility maximization problem, we can restrict attention to the case where the budget constraint holds as an equality. (The argument should not involve the explicit computation of the optimal choices.)

4. Consider the utility function

$$U(x, y) = \min(2x + y, x + 2y)$$

- Draw the indifference curve for $U(x, y) = 20$. Shade the area where $U(x, y) \geq 20$.
- For what values of $\frac{p_x}{p_y}$ will the unique optimum be $x = 0$?
- For what values of $\frac{p_x}{p_y}$ will the unique optimum be $y = 0$?
- If neither x and y is equal to zero, and the optimum is unique, what must be the value of $\frac{x}{y}$?

5. Consider a consumer with a utility function $U(x, y) = e^{(\ln(x)+y)^{1/3}}$.

- State the consumer's utility maximization problem.
- How can you transform the utility function so that this is an easier problem to solve?
- Derive the demand functions. Under what conditions will we get corner solutions for the demand functions?

6. There are two goods, food and clothing, whose quantities are denoted by x and y and prices p_x and p_y respectively. There is a consumer whose income is denoted by I and utility by U . His utility function is

$$U(x, y) = \sqrt{xy}.$$

- Find this consumer demand functions.
- Initially $I = 100$, $p_x = 1$ and $p_y = 1$. What quantities does the consumer buy, and what is his resulting utility?
- Now the price of food rises to $p_x = 1.21$, while income and the price of clothing are as before. What quantities does the consumer buy and what is his resulting utility?

- (d) Suppose the increase in the price of food was caused by the government levying a tax of 21% on food. What is the government revenue from this tax? **Hint: At the new prices, calculate the optimal consumption bundles (x^*, y^*) . Then calculate $(1.21 - 1)x^*$.**
- (e) If the government wants to compensate the consumer by giving him some extra income, how much extra income would be needed to allow him to purchase his original consumption bundle. Is the government's revenue from the tax on the good itself sufficient to provide this compensation? What is the economic intuition of your answer? What quantities will the consumer actually buy? What will be his resulting utility?
- (f) What is the substitution effect induced by the tax? What is the income effect? What is the total effect? Provide a numerical answer and also provide a graphical illustration.
7. As we will see in Wednesday's lecture, Giffen behavior is "paradoxical" in the sense that it implies downward sloping demand. However, it turns out that this behavior can be found in the presence of activities that are addictive and eventually damage one's health. The purpose of this exercise is to explore this issue. Consider the following utility function:

$$u(x_1, x_2) = \ln(x_1 - 1) - 2 \ln(5 - x_2),$$

where x_1 is consumption of food and x_2 is consumption of alcohol. The utility function is defined for $x_1 > 1$ and $0 \leq x_2 < 5$. The interpretation of this utility function is that the consumer needs to consume at least 1 unit of food to subsist and that consumption of alcohol is constrained to be less than 5 units, say because of the risk of alcohol intoxication. Let p_1 be the price of food and p_2 be the price of alcohol.

- (a) Show that the consumer has positive marginal utility for both goods and therefore, will satisfy his budget constraint with equality. In particular, note that the marginal utility of alcohol increases the more of it is consumed, representing the **addictive behavior** of this good.
- (b) State this consumer's utility maximization problem. You can try to solve it if you are courageous. At an interior optimum, the optimal solutions (x_1^*, x_2^*) must satisfy $x_1^* > 1$ and $0 < x_2^* < 5$.
- (c) Irrespective of whether you were able to solve the consumer's problem or not, here I provide the demands that come from an **interior solution**:

$$x_1^* = 1 - \frac{m - p_1 - 5p_2}{p_1},$$

$$x_2^* = 5 + \frac{2(m - p_1 - 5p_2)}{p_2},$$

assuming

$$m - 5p_2 < p_1 < m - 5p_2 + \frac{5p_2}{2} \quad (\star)$$

Why do we need the condition $(\star)$?

- (d) Show that food is an **inferior good** by computing $\frac{\partial x_1^*}{\partial m}$ and determining its sign.
- (e) Compute $\frac{\partial x_1^*}{\partial p_1}$. What is its sign? Under what condition on prices and income does the ordinary demand curve for food is **positively sloped**?
- (f) Show that alcohol is a complementary good to food by computing $\frac{\partial x_2^*}{\partial p_1}$ and determining its sign.

- (g) Let $m = 10$ and the price of alcohol be $p_2 = 1$. Suppose that we start at a price for good of $p_1 = 7$ and then it decreases to $p_1 = 6$. Compute the **substitution**, **income** and **total effects** of such a price decrease. What sign is the substitution effect? What sign is the income effect? What sign is the total effect? Does this corroborate the results from the previous items?
- (h) Putting everything together would yield the following explanation of why the demand for food exhibits Giffen behavior: When the income level is high enough, as is also the pleasure from hazardous drinking, a price fall of food may cause an income effect that stimulates hazardous drinking so much that this crowds out consumption of food. Can you think of other examples in which this explanation makes sense? Provide some other examples.